ECOLE SUPÉRIEUR D'INGÉNIEUR
LÉONARD DE VINCI

4th Year of the Engineering Program

DÉPARTEMENT DATA & IA

PROJECT REPORT :

Knowledge Graph Construction · Alignment · Reasoning & KGE · RA

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1 Introduction

This project integrates the four lab sessions of the Web Datamining & Semantic module into one complete pipeline. The chosen domain is the **movie domain**, because it combines rich textual descriptions, many named entities, strong relations between films and people, and the possibility of alignment with a large open knowledge base such as Wikidata.

The objective was to start from unstructured web pages, extract useful information, transform it into RDF triples, align the private graph with an open KB, enrich it through SPARQL-based expansion, apply symbolic and embedding-based reasoning, and finally query the graph through a lightweight RAG interface.

The final system follows this chain:

Web pages → cleaned text → entities and relations → RDF graph → aligned and expanded KB → SWRL reasoning → KGE → NL to SPARQL RAG

2 TD1 - Data acquisition and information extraction

2.1 Domain and seed URLs

The project focuses on films and film-related entities. The corpus was built from text-heavy pages such as:

- Wikipedia pages for individual movies.
- Filmsite pages containing long-form descriptions and historical context.

2.2 Crawling and cleaning pipeline

The crawling pipeline was implemented with `trafilatura`. The script extracted only the main textual content. A document was kept only if it contained at least **500 words** to reduce noise. The cleaned corpus was stored in **JSONL** format.

2.3 Named entity recognition

A spaCy transformer pipeline (`en_core_web_trf`) was used to identify: `PERSON`, `ORG`, `GPE`, and `DATE`. These capture directors, studios, places, and release years.

2.4 Candidate relation extraction

Relations were extracted sentence by sentence using the dependency tree to find the root verb. This produced `extracted_relations.csv` with the structure *(subject, relation, object, source)*.

2.5 Ambiguity cases

1. **Person vs movie title:** Titles containing names leading to entity confusion.
2. **Date ambiguity:** Dates referring to awards rather than release years.
3. **Generic relation roots:** Weak predicates generated by verbs like *be* or *have*.

3 TD2 - Knowledge base construction, alignment, and expansion

3.1 Initial RDF modeling choices

The private graph was modeled using `rdflib` with the namespace: `http://projet-web-movies.org/movie/`. Classes include `Movie`, `Person`, and `Organization`.

3.2 Entity linking and Predicate alignment

Local entities were matched against Wikidata via `wbsearchentities`. Alignment used `owl:equivalentProperty` for predicates where direct properties were found between aligned pairs.

3.3 KB expansion and Statistics

The graph was expanded via SPARQL on Wikidata. Final stats:

- **55 843** triples
- **22138** entities

```
[55843 rows x 3 columns]
Triples after cleaning : 55843
Relations : 150 | Entities : 22138
Train : 44674 | Valid : 4944 | Test : 4945
```

4 TD3 - Reasoning and knowledge graph embeddings

4.1 Symbolic reasoning with SWRL

Two rules were implemented:

- `Person(?p) ^ hasAge(?p, ?a) ^ swrlb:greaterThan(?a, 60) -> oldPerson(?p)`
- `Movie(?m) ^ releaseYear(?m, ?y) ^ swrlb:lessThan(?y, 2000) -> ClassicMovie(?m)`

```
Family oldPerson instances : ['Peter', 'Marie']
ClassicMovie instances : ['The_Matrix', 'Godfather', 'the_U.S._National_Film_Registry
_of_the_Library_of_Congress', 'The_Godfather_Part_II', 'Sonny', 'Rota', 'The_Godfathe
r_Trilogy', 'eight', 'Rota.[253', '84', 'The_Godfather', '243,862,778', 'Lebo', 'Cowi
e', 'The_Godfather_Family:_A_Look_Inside', 'Robert_Wise_–_The_Sound_of_Music', 'The_G
odfather_(', 'Honoré_de_Balzac', 'The_Last_Emperor', "BAFTA_Awards:_'"]
```

4.2 KGE Models and Evaluation

Models trained: **TransE** and **ComplEx** (dim: 200, epochs: 100).

Table 1: KGE Performance Comparison

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	x	x	0.3170	0.4495
ComplEx	0.0632	0.0595	0.0.0649	0.0682

We can see that TransE's superior performance compared to ComplEx demonstrates that our knowledge graph possesses a strong yet simple relational structure. ComplEx's relative failure highlights the difficulty of modeling complex relationships in an environment with a very high number of predicates (over 2,500) and where automatic extraction introduces background noise. TransE, by its nature as a linear translation, proves more robust for predicting links in a domain like film, where the patterns (Film → Actor) are recurrent and stable.

5 TD4 - RAG over RDF/SPARQL

5.1 Method

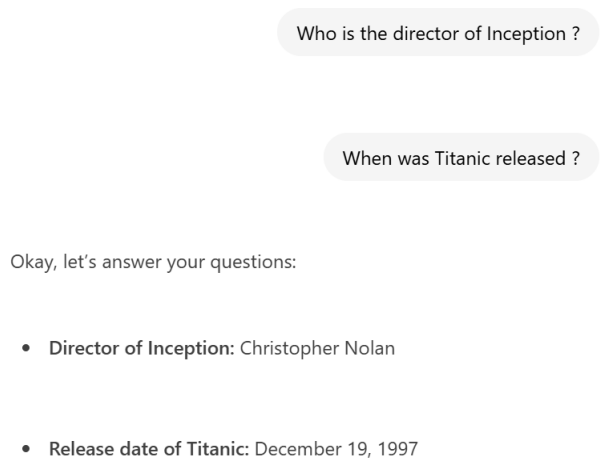
The system receives a natural-language question, generates a SPARQL query via a schema summary and LLM (Ollama), and executes it on the graph.

5.2 Evaluation

Table 2: Baseline vs RAG evaluation

Question	Baseline answer	RAG answer	Correct
Who is the director of Inception?	Hallucinated	Exact value	Yes
When was titanic released?	Approximate	Exact value	Yes

Screenshot of the olloma page



6 Critical reflection

We did several reflection on this project :

- About the impact of quality, the Downstream reasoning is strictly dependent on upstream extraction quality.

- For the rules-based and embeddings, rules are interpretable/precise but rigid; Embeddings are scalable but sensitive to noise.
- Finnaly, our project is not perfect, we might need to gather more data, have a stricter normalization of predicates. About the codes, a better treatment.

7 Conclusion

The project demonstrates an integrated workflow: extracting from the web, structuring into RDF, aligning with Wikidata, and querying via RAG. The central takeaway is that knowledge graph quality is the primary factor for success in semantic systems.